STACKS, DERIVED CATEGORIES OF SPACES: AFFINE-COHERATOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2440-2455 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 2457-2482, the complete author proof (24 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 16 of queue_spaceperfect.json (pb=995, stmt 2440-2455, proof 2457-2482).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. gate_ok=true: raw-interval guard false positive (public-items.jsonl upstream-frame coords); precard_gate.py shift-corrected check cleared this label 2026-10-14 (gate_sp01.sh).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 08F1: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 509–513. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 071Q: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 402–407. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08D9: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2482–2482. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let X be an affine algebraic space over S. Set $A = \Gamma(X, \mathcal{O}_X)$. Then*

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) $Q_X : Mod(\mathcal{O}_X) \to QCoh(\mathcal{O}_X)$ *is the functor which sends* $\mathcal{F}$ *to the quasi-coherent* $\mathcal{O}_X$*-module associated to the* A*-module* $\Gamma(X, \mathcal{F})$*,*
(2) $RQ_X : D(\mathcal{O}_X) \to D(QCoh(\mathcal{O}_X))$ *is the functor which sends* E *to the complex of quasi-coherent* $\mathcal{O}_X$*-modules associated to the object* $R\Gamma(X, E)$ *of* $D(A)$*,*
(3) *restricted to* $D_{QCoh}(\mathcal{O}_X)$ *the functor* RQ_X *defines a quasi-inverse to (08F1).*

Proof. Let $X_0 = \mathrm{Spec}(A)$ be the affine scheme representing X. Recall that there is a morphism of ringed sites $\epsilon : X_{étale} \to X_{0,Zar}$ which induces equivalences

$$QCoh(\mathcal{O}_X) \underset{\epsilon^*}{\overset{\epsilon_*}{\rightleftarrows}} QCoh(\mathcal{O}_{X_0})$$

see Lemma 071Q. Hence we see that $Q_X = \epsilon^* \circ Q_{X_0} \circ \epsilon_*$ by uniqueness of adjoint functors. Hence (1) follows from the description of Q_{X_0} in Derived Categories of Schemes, Lemma 08D9 and the fact that $\Gamma(X_0, \epsilon_*\mathcal{F}) = \Gamma(X, \mathcal{F})$. Part (2) follows from (1) and the fact that the functor from A-modules to quasi-coherent $\mathcal{O}_X$-modules is exact. The third assertion now follows from the result for schemes (Derived Categories of Schemes, Lemma 08D9) and Lemma 071Q. □